

Zach Estrella

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WORK EXPERIENCE

Ziaplex Inc. | Pasig City

Aug 2022 - Oct 2025

Web Developer

Tech stack: PHP (Laravel), JavaScript, MySQL

- Led end-to-end development of 5+ Laravel-based web and back-office systems and contributed to 10+ internal enterprise applications across finance, logistics, and healthcare, supporting 100+ daily users.
- Designed and implemented RESTful APIs and system integrations to enable cross-platform communication, improving data flow efficiency and reducing manual reconciliation by up to 40%.
- Integrated 20+ internal services and proprietary APIs, including webhook-based payment system integrations, improving transaction reliability and reducing operational delays.
- Refactored and stabilized legacy systems by resolving recurring defects and improving architecture, reducing user-reported issues by 30% and increasing system stability.
- Optimized backend performance through MySQL query tuning, Role-Based Access Control (RBAC) implementation, and secure authentication systems, while collaborating with cross-functional teams to deliver scalable solutions across the full Software Development Life Cycle (SDLC).

SKILLS

Languages: PHP, Python, JavaScript, TypeScript

Frameworks: Laravel, Next.js, React

Backend Development: REST APIs, Authentication & Authorization, RBAC, System Integration

Frontend Development: HTML5, CSS3, Tailwind CSS, SEO Optimization

Databases: MySQL, Supabase

Tools & Platforms: Git, Postman, JIRA

AI & Integrations: LLM Integration, Prompt Engineering, Retrieval-Augmented Generation (RAG)

EDUCATION

De La Salle - College of Saint Benilde

Oct 2022

Bachelor of Science in Information Systems

PROJECTS

1. Chatbot

Tech stack: Tailwind, TypeScript, NextJS, LLM integration, Retrieval Augmented Generation (RAG)

- Developed a Retrieval-Augmented Generation (RAG)-inspired chatbot system using a custom-built knowledge base for foundational NLP learning and information retrieval.
- Implemented text embedding generation using Google Gemini Embeddings API, with manually structured and indexed document chunks.
- Built a semantic search and ranking pipeline using cosine similarity and type-aware matching to retrieve the most relevant context from stored embeddings.
- Engineered an LLM-powered response generation pipeline (Google Gemini) by injecting retrieved context into prompts to produce accurate, context-grounded answers.

2. Resume Analyzer

Tech stack: Tailwind, TypeScript, NextJS, LLM integration, Prompt Engineering

- Developed an AI-powered resume optimization platform that helps users tailor resumes for specific job applications using LLM-based feedback and career-focused analysis.
- Implemented resume parsing and job description analysis pipelines to extract key information and generate keyword-matching insights and improvement suggestions.
- Built an interactive AI chat system to support iterative resume refinement, providing contextual feedback and career guidance using LLMs.
- Generated downloadable, ATS-friendly PDF resumes with dynamically structured content and automated formatting for improved readability and recruiter alignment.